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Florida Retail Store Sales Analysis

END-TO-END DATA ANALYSIS USING GOOGLE BIGQUERY, SQL & POWER BI

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1. Executive Summary

This project analyzes two years of retail sales data to identify trends in customer behavior, outlet performance, and the influence of weather conditions on sales. Multiple datasets were integrated using SQL in Google BigQuery, and an interactive Power BI dashboard was developed to support business decision-making.

2. Business Problem

Retail organizations generate large volumes of sales, customer, and operational data every day. When this information is stored across separate datasets, it becomes difficult to understand the factors driving business performance and customer demand.

This project addresses that challenge by integrating retail sales, customer survey, and weather data into a unified analytical dataset using Google BigQuery and SQL. An interactive Power BI dashboard was then developed to enable stakeholders to monitor sales performance, compare store performance, analyze customer demographics, and evaluate the impact of weather conditions on sales.

The objective is to provide actionable insights that support data-driven decision-making in inventory management, workforce planning, promotional campaigns, and overall business strategy.

3. Analysis Objectives

- Integrate retail sales, customer survey, and weather datasets into a single analytics-ready dataset using Google BigQuery and SQL.
- Analyze sales trends across different retail stores, time periods, and customer segments.
- Evaluate the impact of weather conditions, including temperature and rainfall, on daily sales performance.
- Compare store performance using key metrics such as total sales, customer count, and average sales per customer.
- Identify customer demographic patterns by analyzing gender and family versus single customer distributions.
- Develop an interactive Power BI dashboard to visualize key performance indicators (KPIs) and enable data-driven business decisions.

4. Dataset Description

This project combines three datasets representing different aspects of retail operations. These datasets were integrated using **Google BigQuery** and **SQL** to create a single analytics-ready dataset for visualization in **Power BI**.

Dataset	Description	Key Fields
Sales	Contains daily sales transactions for four retail stores in Florida.	Date, Shop ID, Shop Name, Customers, Sales (USD)
Customer Survey	Contains customer demographic information collected through surveys.	Date, % Male, % Female, % Single, % Family
Weather	Contains daily weather observations for the retail locations.	Date, Average Temperature (°F), Humidity, Rain Indicator, Precipitation

5. Tools & Technologies

This project leverages a combination of cloud-based data warehousing, SQL, business intelligence, and version control tools to perform end-to-end data analysis and visualization.

Tool / Technology	Purpose
Microsoft Excel	Source of the raw sales, customer survey, and weather datasets.
Google BigQuery	Imported datasets, stored data, and performed SQL-based data transformation and integration.
SQL (GoogleSQL)	Joined multiple datasets, created derived columns, and prepared the final analytics dataset.
Microsoft Power BI	Designed interactive dashboards and visualized key business metrics and insights.
Visual Studio Code	Managed SQL scripts, project documentation, and repository structure.
Git	Tracked project changes through version control.
GitHub	Hosted the project repository and documentation for portfolio presentation.

6. Project Workflow

This project follows an end-to-end analytics workflow, starting from raw data collection and ending with an interactive business intelligence dashboard. The process involved data ingestion, transformation, integration, visualization, and reporting.

```
                Raw Excel Files
                       
        ┌────────────┬────────────┬────────────┐
        │            │            │
        ▼            ▼            ▼
    Sales Data   Survey Data   Weather Data
        │            │            │
        └────────────┴────────────┘
                     │
                     ▼
          Google BigQuery
      (Data Import & Storage)
                     │
                     ▼
        SQL Data Processing
      • Data Exploration
      • Table Joins
      • Feature Engineering
      • Derived Columns
                     │
                     ▼
      Final Analytics Dataset
                     │
                     ▼
          Export to Excel
                     │
                     ▼
          Microsoft Power BI
      • Data Modeling
      • KPI Development
      • Interactive Dashboard
                     │
                     ▼
          Business Insights
```

6.1. End-to-End Data Pipeline

Stage	Output
Data Collection	Raw Excel datasets
Data Storage	Google BigQuery tables
Data Transformation	SQL-based integrated dataset
Data Export	Analytics-ready Excel file
Data Visualization	Interactive Power BI dashboard
Business Outcome	Actionable insights for decision-making

7. Data Preparation

Before performing the analysis, the raw datasets were imported into **Google BigQuery** and prepared using SQL. The goal was to transform the data into a clean, integrated, and analytics-ready dataset that could be used for visualization in Microsoft Power BI.

7.1. Data Import

Three Excel datasets were imported into Google BigQuery as separate tables:

- **Sales** – Daily sales transactions for each retail store.
- **Survey** – Customer demographic information.
- **Weather** – Daily weather observations.

Each dataset was validated to ensure the data types and column names were correctly imported.

7.2. Data Integration

The datasets were combined using **LEFT JOIN** operations with the **Date** column as the common key. This created a unified dataset containing sales, customer demographics, and weather information.

```
FROM `Florida_Retail_Store.sales` s
LEFT JOIN `Florida_Retail_Store.survey` su
ON s.date = su.date
LEFT JOIN `Florida_Retail_Store.weather` w
ON s.date = w.date
```

7.3. Feature Engineering

Additional columns were created to simplify business analysis and improve dashboard usability.

Derived Column	**Purpose**
Day of Sale	Displays the weekday name (e.g., Monday, Tuesday).
Week of Year	Groups sales by ISO week number.
Year of Sales	Enables year-wise comparisons.
Week Type	Classifies records as **Weekday** or **Weekend**.
Average Sales per Customer	Calculates the average revenue generated per customer.

8. SQL Analysis

SQL played a central role in transforming and integrating the raw datasets into a single analytics-ready table. The analysis involved exploring the data, joining multiple datasets, creating derived fields, and generating business metrics that were later visualized in Power BI.

8.1. Business Metrics

The transformed dataset enabled the calculation of several business metrics used in the dashboard.

Metric	**Purpose**
Total Sales	Measure revenue generated by each store
Customer Count	Analyze store footfall
Average Sales per Customer	Evaluate customer spending
Temperature	Measure weather impact
Rain Indicator	Compare rainy vs. non-rainy days
Customer Demographics	Analyze gender and family distribution
Weekday vs. Weekend	Compare shopping behavior

8.2. SQL Scripts Used

File	**Description**
01_explore_data.sql	Explored the raw datasets
02_data_join.sql	Joined sales, survey, and weather tables
03_create_final_table.sql	Built the final analytics-ready dataset
04_analysis_queries.sql	Generated business metrics and KPI calculations

9. Dashboard Design

The dashboard was designed to provide a clear and interactive view of retail sales performance across four Florida store locations. It enables users to monitor key performance indicators, identify sales trends, compare store performance, and analyze customer demographics and weather conditions through interactive visualizations.

The layout follows a business-oriented design, allowing decision-makers to quickly access important metrics while providing filtering options for deeper analysis.

9.1. Dashboard Objectives

The dashboard was developed to help stakeholders:

- Monitor overall retail sales performance.
- Compare sales across different store locations.
- Analyze customer demographics and purchasing behavior.
- Evaluate the impact of weather conditions on daily sales.
- Explore sales trends across different time periods.
- Support data-driven business decision-making through interactive visualizations.

9.2. Dashboard Components

KPI Metrics

The dashboard includes key performance indicators that provide a quick overview of business performance.

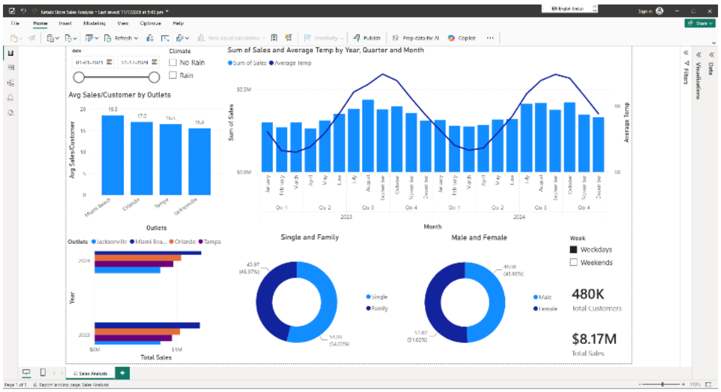
- Total Sales
- Total Customers
- Average Sales per Customer
- Average Temperature

These KPIs help users understand the overall business performance at a glance.

9.3. Dashboard Design Principles

The dashboard was designed to provide business users with a clear and interactive view of retail sales performance across four Florida store locations.

- Simple and consistent layout
- Clear chart titles and labels
- Interactive filtering for easy exploration
- Appropriate chart selection for different data types
- Minimal visual clutter to improve readability



10. Key Business Insights

The dashboard provides several insights into sales performance, customer behavior, and the influence of external factors on retail operations. The following observations summarize the key findings from the analysis.

10.1. Store Performance

Among the four retail stores, sales performance varied significantly. Miami Beach generated the highest average sales per customer, averaging **$18.5**, while Jacksonville recorded the lowest at **$15.6**.

Business Impact

High-performing stores can serve as benchmarks for operational practices, while lower-performing stores may require targeted marketing campaigns or operational improvements.

10.2. Sales Trend Over Time

The sales trend shows noticeable fluctuations throughout the two-year period, suggesting the presence of seasonal demand patterns rather than consistent daily performance.

Business Impact

Understanding these trends helps management improve inventory planning, staffing schedules, and promotional timing during high-demand periods.

10.3. Customer Demographics

The customer demographic analysis indicates a balanced distribution between male and female customers. The dashboard also highlights the proportion of family and single shoppers, providing valuable insights into the primary customer segments.

Business Impact

These insights can support targeted marketing strategies and promotional campaigns based on customer preferences.

10.4. Average Sales per Customer

The calculated average sales per customer provides a useful measure of customer spending across stores. Comparing this metric helps identify locations where customers tend to spend more per visit.

Business Impact

Stores with higher average spending may offer opportunities to replicate successful product placement or pricing strategies across other locations.

10.5. Weather Influence on Sales

By combining weather and sales data, the dashboard enables analysis of how environmental conditions may influence customer traffic and purchasing behavior.

While weather alone does not determine sales performance, it provides additional context that can support operational planning.

Business Impact

Weather forecasts can be incorporated into staffing, inventory allocation, and promotional planning to improve operational efficiency.

10.6. Weekday vs. Weekend Performance

The analysis allows comparison of sales between weekdays and weekends, helping identify periods of increased customer activity.

Business Impact

Understanding shopping patterns enables better workforce scheduling and resource allocation during peak business hours.

10.7. Interactive Business Monitoring

The dashboard provides dynamic filtering by date, weather conditions, and day type, allowing users to explore sales performance from multiple perspectives without modifying the underlying data.

Business Impact

Interactive reporting enables business users to quickly answer operational questions and make informed decisions based on current business needs.

Summary of Insights

Insight	**Business Value**
Store Performance Comparison	Identifies high- and low-performing stores
Sales Trend Analysis	Supports seasonal planning and forecasting
Customer Demographics	Enables targeted marketing strategies
Average Sales per Customer	Measures customer spending behavior
Weather Analysis	Provides context for operational planning
Weekday vs. Weekend Analysis	Improves staffing and inventory planning
Interactive Dashboard	Supports data-driven decision-making

11. Future Improvements

Based on the insights generated from this analysis, several opportunities exist to enhance both the analytics solution and business operations.

11.1. Optimize Staffing and Inventory

Leverage seasonal sales trends and weather forecasts to improve workforce scheduling and inventory planning during high-demand periods.

Expected Benefit

- Better product availability

- Reduced operational costs
- Improved customer experience

11.2. Implement Weather-Based Promotions

Introduce targeted promotional campaigns during rainy days or periods of lower customer traffic to maintain consistent sales.

Expected Benefit

- Increased customer footfall
- Higher sales during low-demand periods

11.3. Expand High-Performing Store Operations

Analyze successful stores to identify best practices and replicate them across other retail locations.

Expected Benefit

- Improved overall store performance
- More consistent customer experience

11.4. Develop Customer Loyalty Programs

Use customer demographic insights to design loyalty and retention programs tailored to different customer segments.

Expected Benefit

- Increased repeat customers
- Higher customer lifetime value

11.5. Personalize Marketing Campaigns

Develop targeted marketing strategies based on customer demographics and shopping behavior.

Examples include:

- Weekend promotions for families
- Weekday offers for individual shoppers
- Location-specific campaigns based on customer demand

Expected Benefit

- Higher campaign effectiveness
- Improved customer engagement

11.6. Enhance the Analytics Platform

Future iterations of this solution could include the following enhancements:

- Connecting Power BI directly to Google BigQuery
- Automating dashboard refreshes

- Adding executive KPI dashboards
- Incorporating additional business data such as inventory, product categories, and promotions
- Expanding the solution to support real-time business monitoring

11.7. Future Roadmap

Area	Future Enhancement
Business Operations	Optimize staffing and inventory planning
Marketing	Weather-based and demographic-based promotions
Customer Experience	Loyalty and personalized marketing programs
Analytics	Direct BigQuery–Power BI integration
Reporting	Automated refresh and executive dashboards
Scalability	Integration of additional business datasets

12. Conclusion

This project provided valuable hands-on experience in building an end-to-end analytics solution using Google BigQuery, SQL, and Power BI. Starting with separate sales, customer survey, and weather datasets, I integrated and transformed the data into a single analysis-ready dataset that could be used for meaningful business reporting.

Using this dataset, I developed an interactive Power BI dashboard that helps visualize sales trends, compare store performance, understand customer demographics, and analyze how weather conditions may influence retail sales. The dashboard is designed to help stakeholders quickly explore key metrics and make informed business decisions.

Throughout this project, I strengthened my practical skills in SQL, data transformation, cloud-based data storage, and dashboard design. More importantly, I learned that effective data analysis is not just about creating visualizations; it's about understanding the business problem, preparing reliable data, and presenting insights that support better decision-making.

Overall, this project reflects my ability to work through the complete data analytics process, from collecting and preparing raw data to building an interactive dashboard that delivers actionable business insights. It has provided a strong foundation for tackling more complex analytics projects and further developing my skills as a Data Analyst.